Experiment 1

Aim

Understand the working of following (i) IP Address. (ii) Cisco IOS. (iii) Straight Cable & Cross Cable, RJ45 (iv) Layer 2 Switch. (v) Router.

Objective

1. To understand the concept and significance of IP addressing in computer networks.
2. To study the architecture and working of Cisco IOS along with its different command-line modes.
3. To identify the differences between Straight-Through and Cross-Over Ethernet cables and their applications.
4. To understand the structure and purpose of the RJ45 connector used in Ethernet communication.
5. To study the working principle of a Layer-2 Switch and its role in forwarding data frames using MAC addresses.
6. To understand the functions of a Router and its role in connecting different networks using IP addresses.
7. To gain practical exposure to Cisco Packet Tracer for basic network configuration and simulation.
8. To analyze communication between network devices by configuring IP addresses and verifying connectivity using network utilities such as **ping**.

Practical / Experiment Steps

1. Open **Cisco Packet Tracer** on the computer system.
2. Study the concepts of **IPv4**, **IPv6**, subnet mask, network ID, and host ID.
3. Create a simple network topology by connecting two PCs and assign valid IP addresses.
4. Verify communication between the PCs using the **ping** command.
5. Add a Cisco Router in Packet Tracer and access the **Cisco IOS Command-Line Interface (CLI)**.
6. Explore the different Cisco IOS modes, including **User EXEC Mode**, **Privileged EXEC Mode**, and **Global Configuration Mode**.
7. Execute basic Cisco IOS commands such as **show version**, **show running-config**, **show ip interface brief**, **ping**, and **configure terminal**.
8. Study the construction and applications of **Straight-Through Cable**, **Cross-Over Cable**, and **RJ45 Connector**.
9. Create a network topology using a **Layer-2 Switch** to observe frame forwarding and communication between connected devices.
10. Study the functions and working of a **Router** in connecting different networks and forwarding packets based on IP addresses.
11. Observe the simulation results and verify successful communication between network devices.
12. Record the observations, capture the required screenshots, and save the Packet Tracer file for future reference.

Procedure of the Experiment

1. Start the computer system and launch **Cisco Packet Tracer**.
2. Create a new workspace for the experiment.
3. Place two PCs in the workspace and connect them using the appropriate networking device (direct connection or Layer-2 Switch as required).

4. Assign valid **IP addresses** and **Subnet Masks** to both PCs through the **Desktop → IP Configuration** option.
5. Verify network connectivity by opening the **Command Prompt** and executing the **ping** command between the connected PCs.
6. Add a **Cisco Router** to the topology and access its **Command-Line Interface (CLI)**.
7. Explore the different **Cisco IOS modes** such as **User EXEC Mode**, **Privileged EXEC Mode**, and **Global Configuration Mode**, and execute basic IOS commands.
8. Study the structure and applications of **Straight-Through Cable**, **Cross-Over Cable**, and the **RJ45 Connector** used in Ethernet networking.
9. Create a network topology using a **Layer-2 Switch** and observe how it forwards frames based on **MAC addresses**.
10. Study the working of a **Router** by understanding how it forwards packets between different networks using **IP addresses** and the routing table.
11. Observe the output, verify successful communication between devices, and analyze the results.
12. Save the Packet Tracer project, record the observations, and attach the necessary screenshots for documentation.

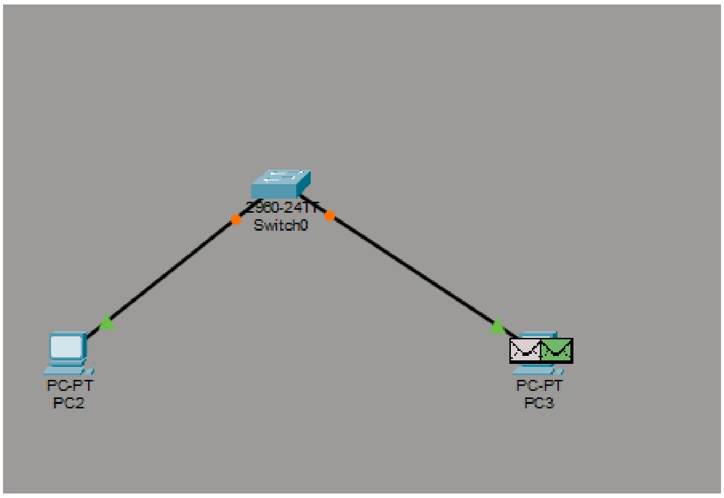
Input / Output Details

Input

Network topology, IP addresses, and Cisco IOS commands entered in Cisco Packet Tracer.

Output

Successful network communication and verification of networking concepts through Cisco Packet Tracer.



Learning Outcome

After completing this experiment, the student was able to understand the concepts of IP addressing, Cisco IOS, Ethernet cables, RJ45 connectors, Layer-2 switches, and routers. The experiment also provided practical experience in configuring basic network devices and verifying communication using Cisco Packet Tracer.